

DHEERAJ REDDY PANDHAM

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PROFESSIONAL SUMMARY

Software engineer since 2021, building Go backend services, APIs and cloud data workflows on AWS. Currently at Capital One on data platform pipelines that **process millions of records daily** and multitenant capabilities serving **five lines of business**. Experience across Kafka stream processing, data ingestion and egress, data quality, event-driven architecture, application security and CI/CD. Set and enforce team testing conventions at **90%+ SonarQube coverage**, review peers' pull requests and onboard newer engineers. AWS Certified Developer, Associate.

EDUCATION

University of Central Missouri

M.S., Computer Science, GPA 3.75 / 4.00

Completed Dec 2023

Missouri, USA

Vidya Jyothi Institute of Technology

B.Tech., Computer Science, GPA 3.82 / 4.00

Apr 2022

Hyderabad, India

CERTIFICATIONS & RECOGNITION

AWS Certified Developer, Associate; Certified Cloud Practitioner

Patent MONSTOGRain, IoT grain-storage monitoring. India patent application 202041049928A, published Nov 16, 2020.

Other Front-End Web Development with React; Programming Essentials in Python; Cisco Programming Essentials in C

PROFESSIONAL EXPERIENCE

Capital One

Software Engineer, Richmond, VA

Sep 2024 - Present

Go, Kafka, AWS, React

- Develop Go backend services and APIs supporting dataset ingestion, egress and data-quality workflows **processing millions of records daily** across AWS S3, ECS, EMR, Glue, Batch and RDS.
- Build multitenant support across Go backend services and React components serving **five Capital One lines of business**, implementing tenant resolution through the request path and per-tenant feature configuration, and contributing tradeoff analysis on both approaches during design review.
- Build Kafka stream-processing workflows for dataset pipelines and improve data quality across ingestion and egress jobs.
- Build event-driven processing with AWS Step Functions and Lambda, and **redesign SQS error handling from batch to per-message processing and deletion**, so a single failure no longer sent its entire 10-message batch to the dead-letter queue. Eliminated routine manual redrives, duplicate reprocessing and false DLQ-depth pages.
- Identify and remediate application security vulnerabilities in internally developed applications to reduce attack exposure.
- Set and enforce the team's testing conventions at **90%+ SonarQube coverage** across backend services, APIs and data-processing workflows; review peers' pull requests and onboard newer engineers onto the Go services and AWS data workflows.

ADT

Software Engineer, Remote

Dec 2023 - Sep 2024

Go, AWS Serverless, PostgreSQL

- Built and maintained Go backend services powering customer account and self-service features in ADT's residential home-security platform.
- Designed and built REST APIs consumed by ADT's customer-facing web and mobile applications.
- Implemented serverless workflows with AWS Lambda and Step Functions to automate order and installation-scheduling processing, integrating third-party services into those flows.
- Built PostgreSQL data-access logic and optimized database operations to improve query performance across account and order flows.
- Deployed workloads on AWS EC2, S3 and RDS with IAM roles and policies for least-privilege access, and enhanced GitHub Actions CI/CD pipelines.
- Supported production on call, triaging incidents and debugging issues through CloudWatch log analysis and SNS alerting.

Rakuten India

Software Engineer, Bangalore, India

Dec 2021 - Jul 2022

Java, Spring, React, SQL, MongoDB

- Built full-stack features for a consumer movie ticketing portal spanning showtime listings and search, seat selection and booking, checkout and payments, and user accounts with booking history.
- Implemented **seat-hold concurrency handling** to prevent double-booking when multiple users selected the same seat.
- Added caching in front of showtime and listing read paths to reduce repeat query load on high-traffic browsing pages.
- Optimized SQL and MongoDB queries behind the portal, **reducing page load times by 40%** across browsing and booking flows.
- Developed responsive React components for the booking interface and Java services with the Spring Framework, with JUnit and Mockito unit tests covering booking and account logic.
- Improved Jenkins and GitHub Actions pipelines to streamline deployments and reduce manual intervention.

TECHNICAL SKILLS

Languages Go (Golang), Java, Python, JavaScript, SQL, PL/SQL, HTML, CSS, JSON, XML

Backend REST APIs, API development, Gin, Echo, Gorilla Mux, GORM, Spring Boot, Spring MVC, Hibernate, Django, Node.js

Cloud & Data AWS S3, ECS, EMR, Glue, Batch, RDS, Secrets Manager, Neptune, EC2, IAM, Fargate, Lambda, Step Functions, SQS, SNS, CloudWatch; Kafka, PostgreSQL, MySQL, MongoDB, Snowflake

Data Practices Multitenancy, data ingestion, data egress, data quality, stream processing, event-driven architecture, batch processing, caching, query optimization, concurrency control

DevOps & Testing Docker, Kubernetes, Jenkins, GitHub Actions, CI/CD, Testify, JUnit, Mockito, SonarQube, PagerDuty

Security & Process Application security, vulnerability remediation, production support, code review, Agile, Scrum, Git, Jira, Maven, Postman

Frontend React, Next.js, Bootstrap, jQuery, Chart.js

SELECTED PROJECTS

Monstograin: IoT grain-storage monitoring	Patent 202041049928A
• Built an IoT system monitoring temperature and humidity inside food-grain storage facilities, applying machine learning to sensor data to detect anomalies and help prevent spoilage. • Developed React and Chart.js dashboards visualizing storage conditions and operational insights. Python, React, Node.js, Chart.js.	
TradeSense: risk-first algorithmic trading system	Python Live demo
• Built a market-prediction and trading system in Python, Streamlit, Pandas and scikit-learn, with backtesting against real and synthetic datasets to evaluate strategies.	
CKart: regional e-commerce platform	Django, Next.js, K8s Live demo
• Built an e-commerce platform for construction supplies with authentication, address and payment management, inventory tracking, order processing, district-based dynamic pricing and promotional discounts. • Containerized and deployed with Docker, Kubernetes and Jenkins on PostgreSQL.	